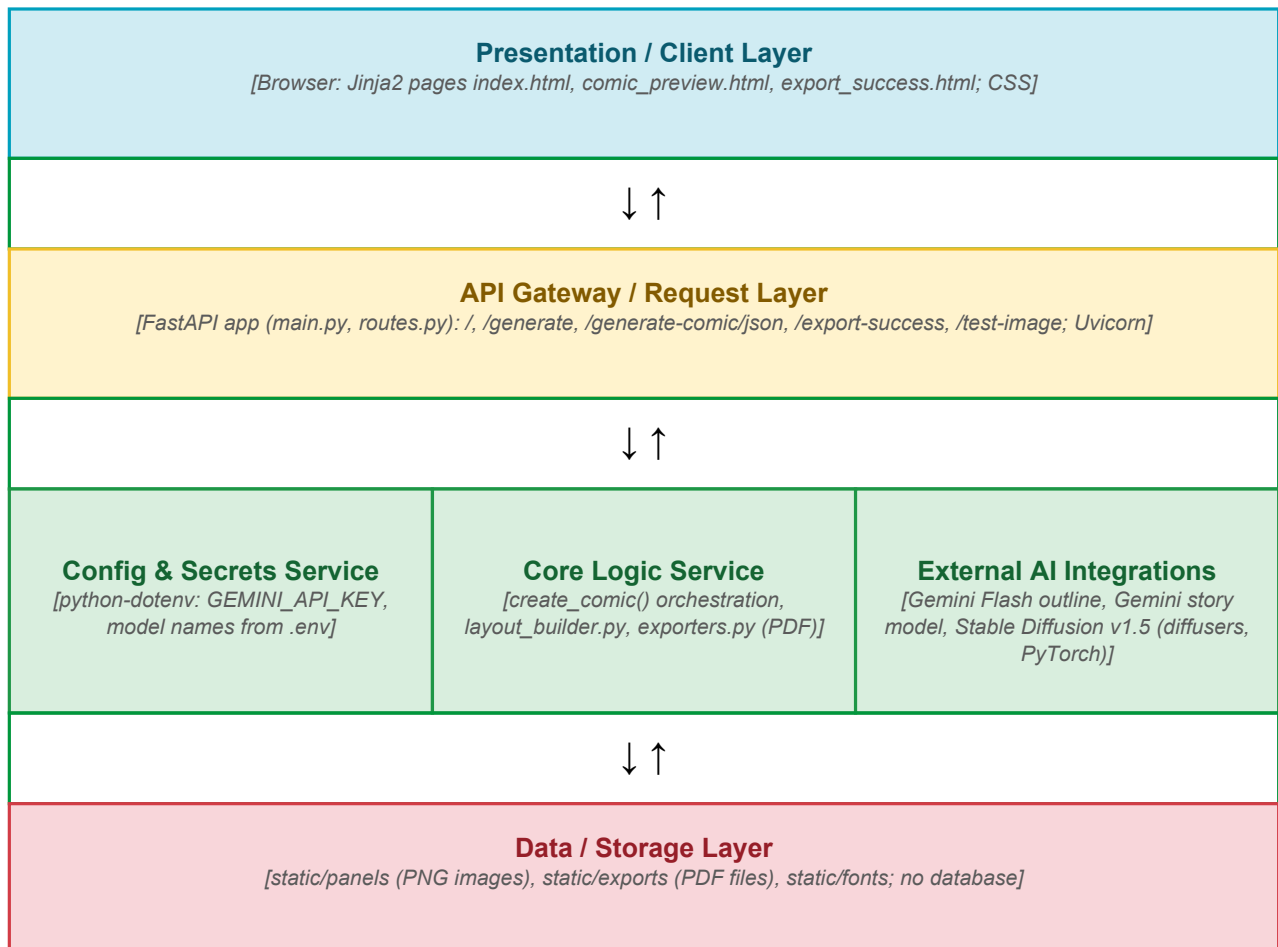


## Solution Architecture

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| <b>Date</b>          | 2 October 2026                                          |
| <b>Team ID</b>       | SWTID-2026-8368                                         |
| <b>Project Name</b>  | ComicCraft - AI Comic Story Creator using Gemini Models |
| <b>Maximum Marks</b> | 5 Marks                                                 |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



### Component Description Table:

| Component Name | Description / Role                                     | Technologies Used |
|----------------|--------------------------------------------------------|-------------------|
| Web UI         | Home form, comic preview and export confirmation pages | Jinja2, HTML, CSS |

|                 |                                                                    |                                   |
|-----------------|--------------------------------------------------------------------|-----------------------------------|
| FastAPI app     | Mounts /static, defines routes, starts model warm-up thread        | FastAPI, Uvicorn                  |
| Outline service | Returns a 5-panel JSON outline (panel, title, scene, image_prompt) | gemini_flash.py, Gemini Flash     |
| Story service   | Writes caption and narration for each panel                        | gemini_pro.py, Gemini             |
| Image service   | Cached Stable Diffusion pipeline; saves PNG; returns its URL       | diffusers, PyTorch (MPS/CUDA/CPU) |
| Layout builder  | Parses the story with regex and merges it with outline and images  | Python (pure function)            |
| Exporter        | Writes a PDF with one page per panel                               | fpdf2, DejaVu fonts               |
| Storage         | Generated PNG panels and PDFs on disk                              | File system                       |